



# **Emotion Dataset for Emotion Recognition Tasks**

**CSE422: Artificial Intelligence**

**Project Report**

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# Introduction

The dataset consists of English-language Twitter messages categorized into six primary emotions: anger, fear, joy, love, sadness, and surprise. Further details about the dataset can be found in the referenced paper. To create the dataset, the authors designed a set of hashtags to extract a distinct collection of English tweets from the Twitter API.

## Dataset Description

The dataset consists of three subsets: **Training**, **Validation**, and **Test**, all structured identically. Each dataset entry represents a snippet of text paired with a numerical label. Here's a detailed overview:

### **Source/Link:**

[Emotion Dataset for Emotion Recognition Tasks](#)

**Reference:** Elvis Saravia, Hsien-Chi Toby Liu, Yen-Hao Huang, Junlin Wu, and Yi-Shin Chen. 2018. [CARER: Contextualized Affect Representations for Emotion Recognition](#). In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 3687–3697, Brussels, Belgium. Association for Computational Linguistics.

### **Features:**

- **Text:** A snippet of text, often representing a statement, thought, or feeling.
- **Label (Emotion):** numerical value (0 to 5), representing a category or sentiment associated with the text.

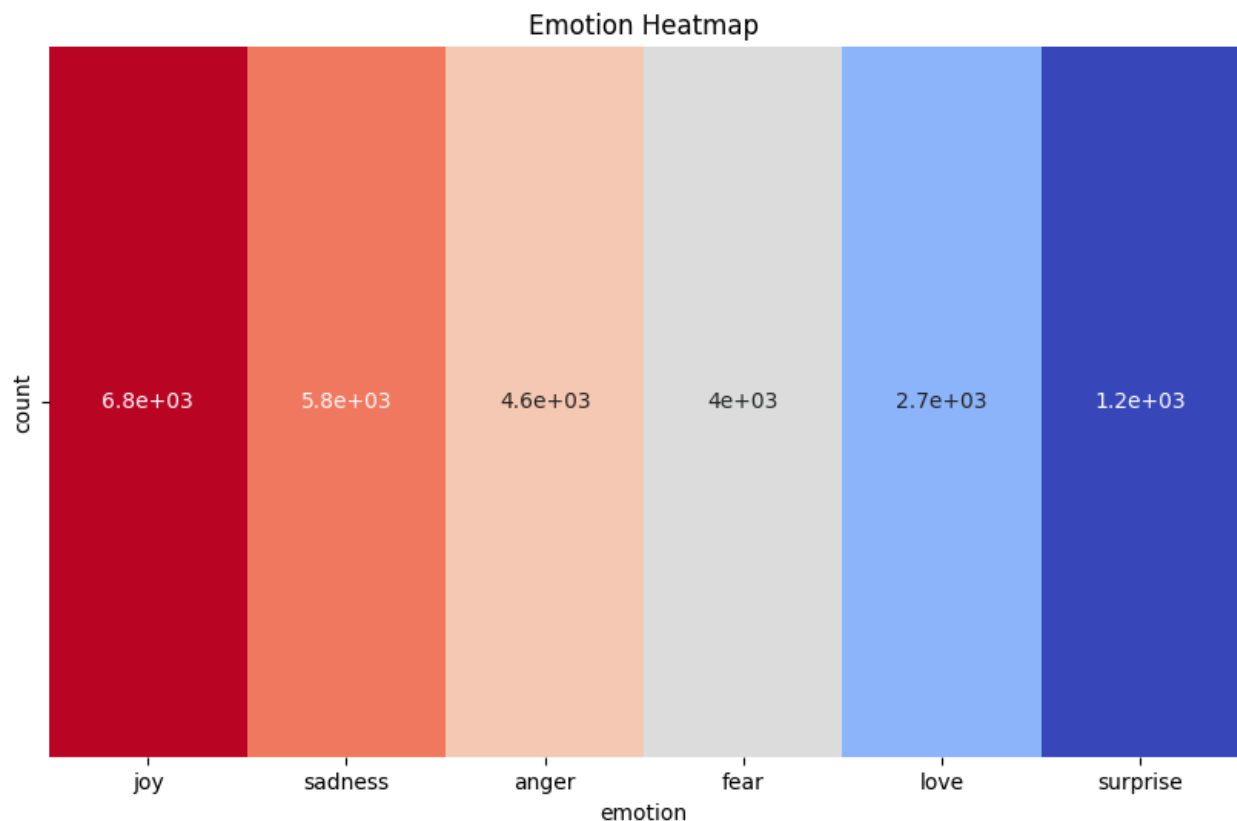
### Dataset Sizes:

- **Training Dataset:** The largest subset, with 16,000 rows, used to train a model.
- **Validation Dataset:** A smaller subset of 2,000 rows, used to tune and evaluate the model during training to prevent overfitting.
- **Test Dataset:** Another subset of 2,000 rows, used to assess the final model's performance on unseen data.

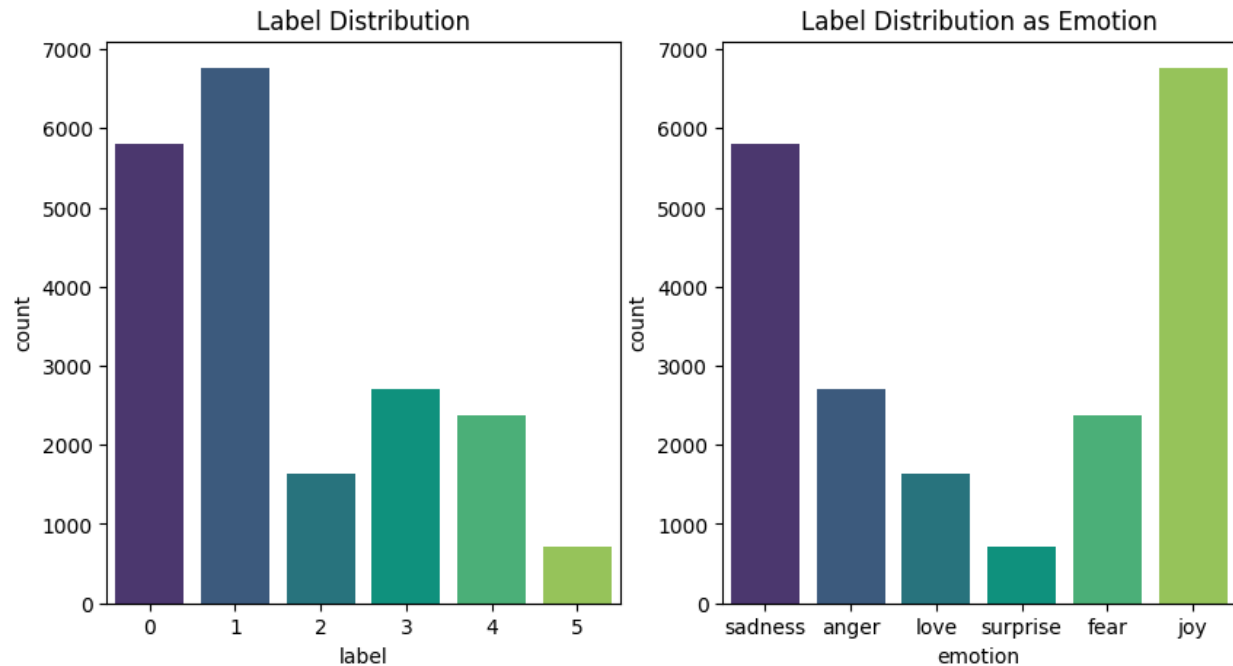
**Problem & feature type:** The "Emotion Dataset for Emotion Recognition Tasks" is typically a **classification** problem. The goal is to classify each tweet into one of the predefined emotion categories. Also, the target variable (emotion) is **categorical**. The model is tasked with predicting the appropriate emotion label for a given tweet.

**Datapoints:** This dataset with its combined subsets contains 20,000 tweets in total.

### Heatmap



## Bar chart to represent all labels



## Dataset Pre-Processing

Initially, we worked with the Train Dataset, but for the final output, we restructured our code and concatenated Training, Test, and Validation into a single Dataset variable, which now has a total of 20,000 Rows. The Dataset being somewhat modern has no NULL values, but a duplicate and we dropped that by applying the **drop\_duplicates()** function.

## Feature Scaling

As there were some labels with significantly lower amounts of values, we had to perform oversampling over those, importing **resample**.

# Dataset Splitting

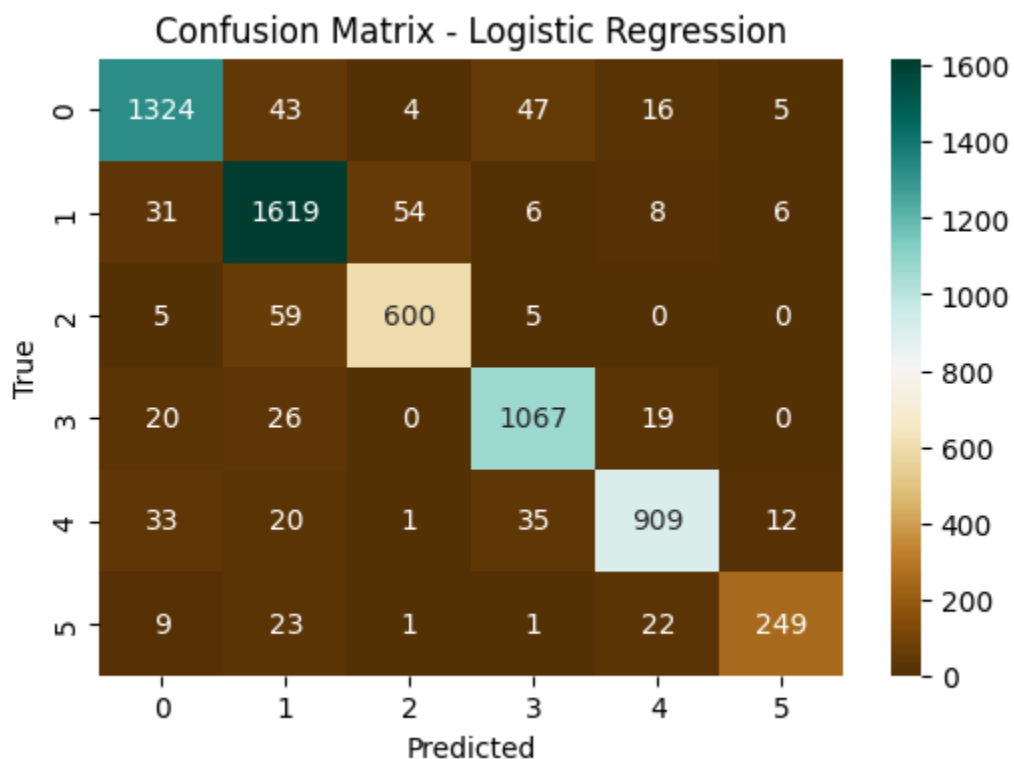
For the final output, we decided to three-way split the data into 70-20-5. This three-way split ensures the following:

- **70% Training:** Provides enough data for the model to learn effectively, especially for larger datasets.
- **20% Test:** Ensures a robust and reliable evaluation, accounting for variability in the data.
- **5% Validation:** Ensures the best settings for the **parameters** of a machine learning model that are not learned during training.

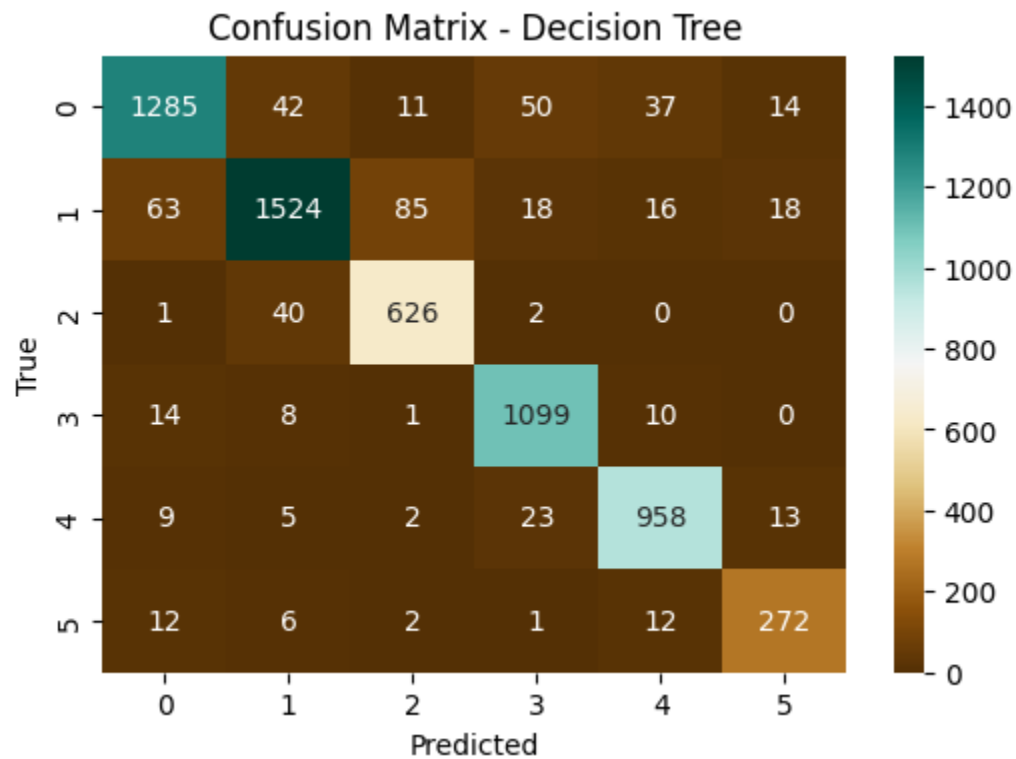
# Model Implementation

For this Dataset we chose to work with, given that it's a **classification** problem, we ran it through the following 5 Models along with their confusion matrix.

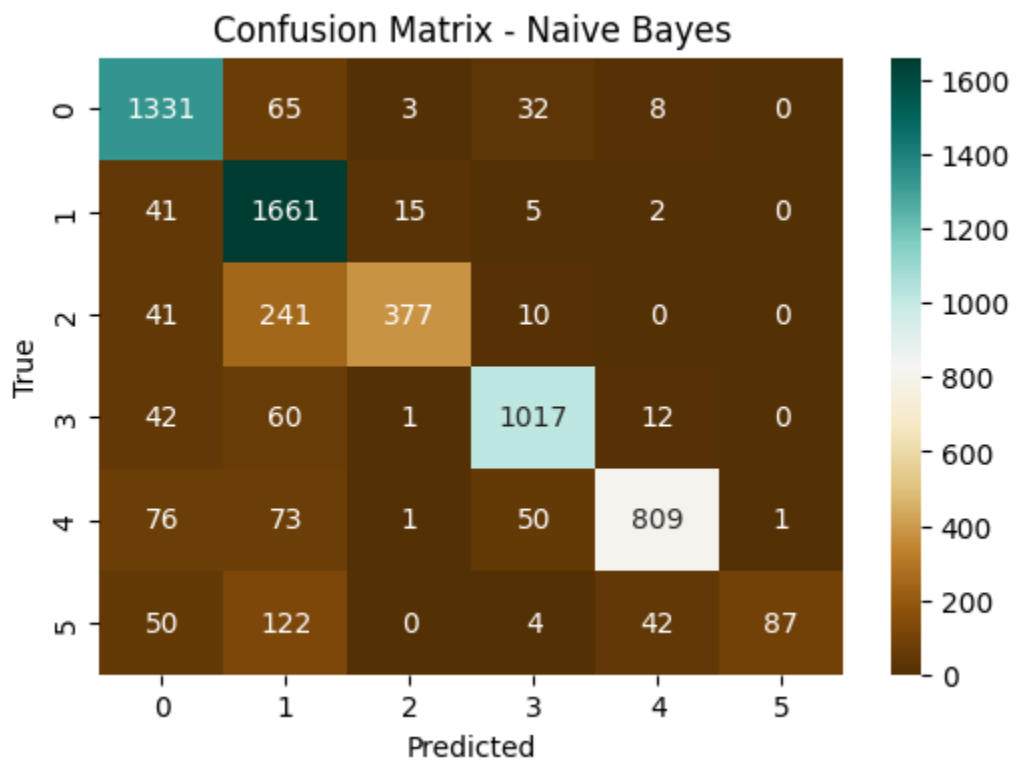
- **Logistics Regression**



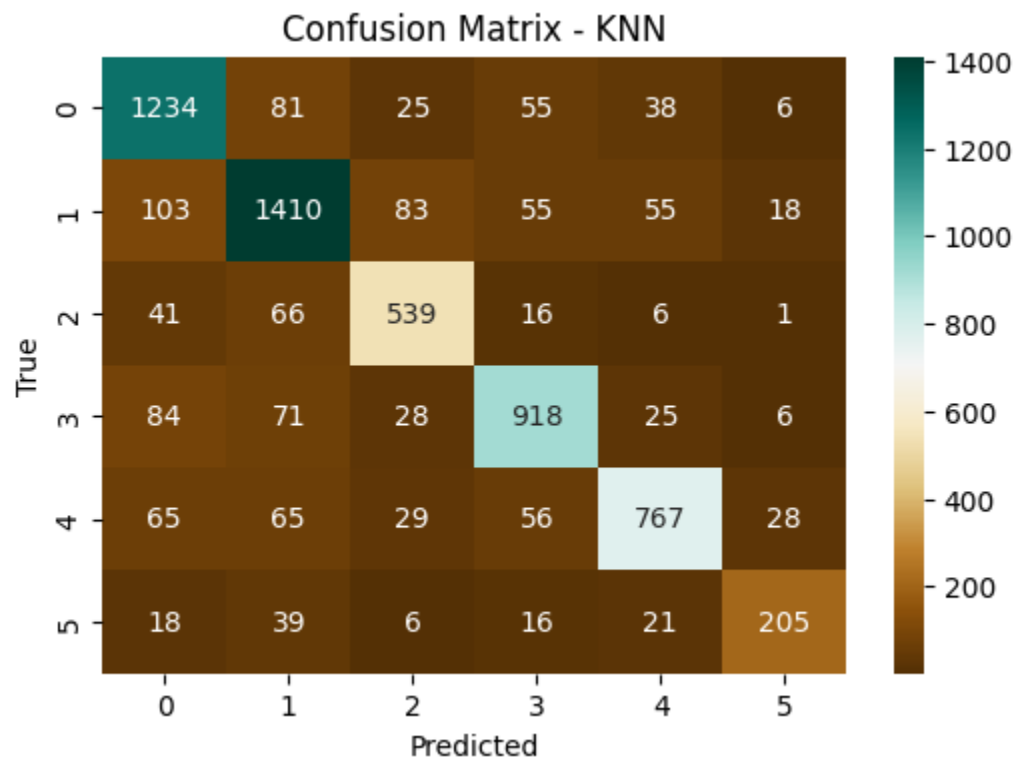
- Decision Trees



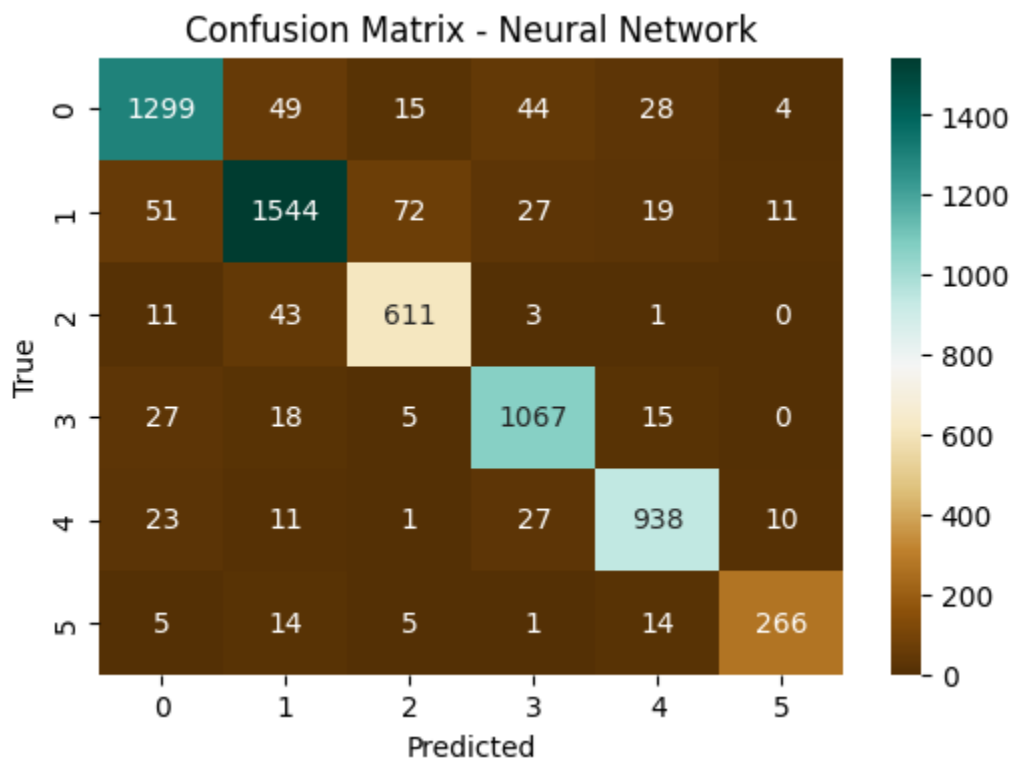
- Naive Bayes



- **KNN**

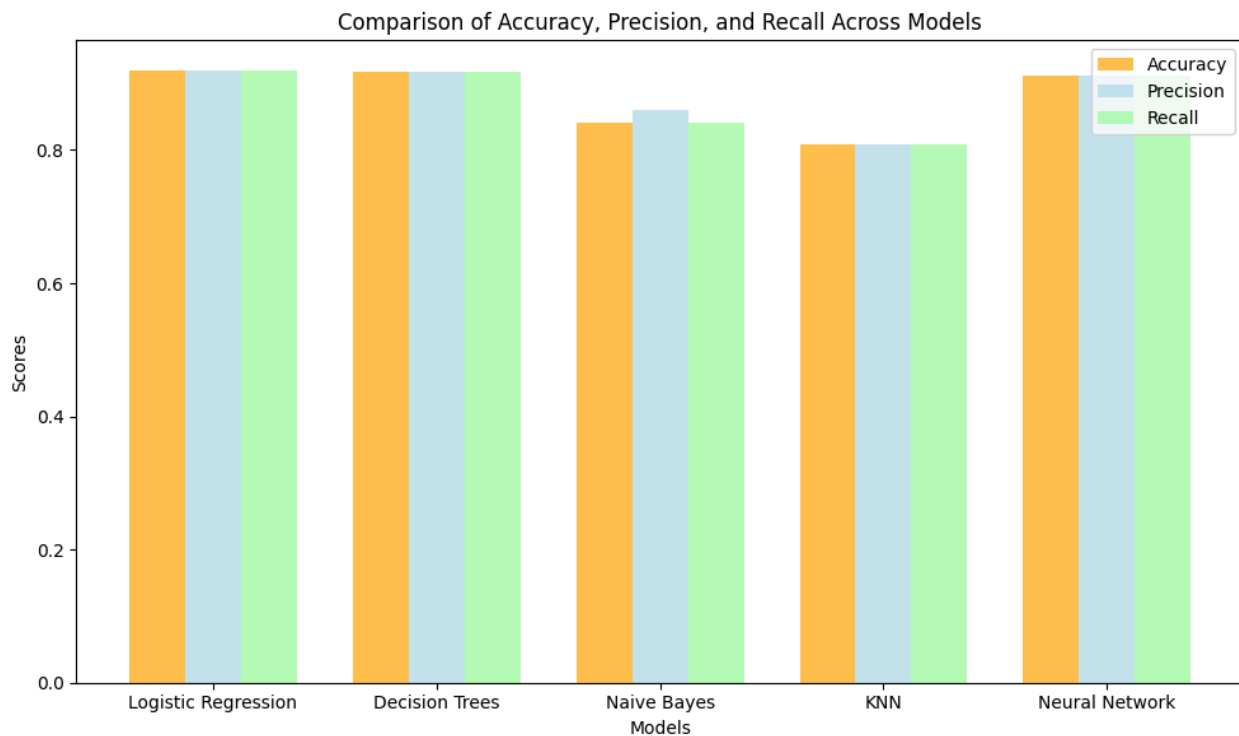


- **Neural Network**





# Analysis/Observations



## 1. **Logistic Regression:**

- Scores are consistent across Accuracy, Precision, and Recall, indicating balanced performance.
- Likely a strong baseline model for our classification task.

## 2. **Decision Trees:**

- Similar performance to Logistic Regression but may have slightly higher variability, suggesting that it performs well but might overfit depending on the tree depth.

## 3. **Naive Bayes:**

- Slightly lower scores across the metrics compared to Logistic Regression and Decision Trees.
- This could be due to assumptions of feature independence, which may not fully align with our dataset.

4. **KNN:**

- Noticeably lower performance compared to the other models.
- KNN might struggle due to high dimensionality or imbalanced data in our dataset.

5. **Neural Network:**

- Appears to have similar performance to Logistic Regression and Decision Trees across all metrics, indicating strong predictive capability.
- Suggests that our dataset benefits from the complexity and representational power of a Neural Network.

## Conclusion

If **performance metrics are similar**, we should choose a model based on our priorities:

- For **simplicity and interpretability**, we can go with either **Logistic Regression** or **Decision Trees**.
- For **future scalability or more complex datasets**, we can use **Neural Network**.

For this Dataset in particular, it's best to go with Logistic Regression or Decision Tree.